A complete reading companion to Anticipating Network Contagions

How to use this document

This companion follows Patrick Iben's *Anticipating Network Contagions: When Directedness Outperforms Connectedness*, dated 5 July 2026, passage by passage. It is for a numerate reader who knows averages and graphs but is new to network control. The concise study guide offers the shorter route. Here, work through the units in order, then use the glossary and answered transfer test.

The exact basis is the author draft, SHA-256 b48f19b5a71943f540728fb197cb5d84aeea782a2aaad5a8b6144fe216755509. Page numbers refer to that sixteen-page PDF. Complete coverage maps its substantive passages, equations and figures to the units below. This is a teaching explanation of that frozen draft, not a new paper or an empirical rerun. Where the draft refers to supplementary analyses not printed in it, this document explains their stated role without inventing their contents.

These boundaries apply throughout:

- The stress series are empirical; the support interventions and cascade reductions are simulated on calibrated, rescaled linear surrogates.
- A predictive transmitter is not an identified physical cause or a potential-outcomes treatment effect.
- Five prediction-first cases lack an independently verified pre-analysis public timestamp. Two of those predictions failed.
- Most individual advantage intervals include zero. Only equity and the held-out COVID-19 network individually exclude it in the reported block bootstrap.
- The raw-data forecasting result is non-significant. It does not validate intervention benefits.
- The geometric account is interpretive. Neither a universal control theorem nor an operational emergency, clinical or financial policy is established.

Evidence labels distinguish **DEFINITION**, **ANALYTIC**, **MODELED_LOCKED** (a specified simulation), **MEASURED_DATED** (analysis of dated observations), **INTERPRETIVE** and **SCOPE**. A unit can contain more than one class; its evidence field keeps them separate. Human audience, accessibility and transfer assessments have not been completed. This document does not certify full G14 or full manuscript validation.

Reading the question and the model

D01. What the abstract promises, and what it actually measures

Where we are: Abstract, page 1; overview diagram, Figure 1, page 14.

What the passage is doing: The abstract announces a comparison between two allocation rules. One supports the unit with the largest realized stress; the other supports the unit diagnosed as exporting

stress-related predictive influence. Seven empirical networks provide the fitted dynamics. The controller experiments then occur inside those fitted models. The abstract also announces the three main verification objects: symmetrization, a prediction-first record and recovery of hypothesized historical origins. They are not three replications of a field intervention.

Worked reasoning: Reason it through: write a chain with three boxes: observations, fitted surrogate, simulated response. A reported cascade reduction belongs to the third box. A claim that Wisconsin has a fitted transmitting role belongs to the middle box. A measured daily pollution value belongs to the first. Moving a statement between boxes changes its evidence. The abstract's strong language about structural leverage should therefore be read with the model qualifier attached, not as a measured benefit of a real response plan.

In ordinary language: A map can help you ask where to place support. But a contest played on that map is still a model experiment. Here the map comes from real records. The test of support takes place in a simplified system built from them. Keep those stages separate whenever you repeat a headline.

Evidence status: Mixed: MEASURED_DATED inputs and fitted diagnostics; MODELED_LOCKED intervention results; SCOPE for transfer. The abstract's approximate recovery probabilities are developed later, not estimates of a policy's success probability.

Do not accidentally conclude: Seven networks mean seven successful interventions. They include different empirical outcomes, null cases, failures of prior domain predictions and uncertain surrogate advantages.

Try it: Rewrite “anticipation reduced real network damage” faithfully. **Answer:** “Transmitter-targeting reduced modeled cascades in specified calibrated surrogates; the draft tests when that advantage appears.” That version preserves both the allocation comparison and its evidence ceiling.

D02. Interference changes the allocation question

Where we are: Introduction opening, page 1; Methods causal-scope statement, pages 9–10.

What the passage is doing: The introduction uses interference to motivate a network problem. Interference means that one unit's outcome can depend on what happens to other units, including their treatment. The no-interference part of SUTVA rules that dependence out. This paper does not estimate a formal causal effect under a treatment-assignment design. It uses the interference literature to explain why deciding where to support a system may require information beyond each unit's own condition.

Worked reasoning: A useful reasoning example needs no invented data. Imagine two connected towns, one receiving a disturbance and one passing it onward. Relief for current suffering and source control answer different questions. A policy could need both. The paper compares stylized resource-allocation rules under a fixed model budget; it does not assign moral value to the towns. Its “loudest” variable is the modeled stress measure, not a survey of clinical need, vulnerability or total harm.

In ordinary language: Helping one place may change what happens elsewhere. That makes a list of local damage incomplete for deciding how to limit spread. You may need to know which places pass stress onward. Still, knowing that spillovers are possible does not tell you how large a real intervention effect will be.

Evidence status: DEFINITION for interference and the operational decision; SCOPE for causal interpretation. The references to partial and cross-cluster interference are background, not a claim that this analysis implements those identification methods.

Do not accidentally conclude: A source ranking licenses withholding help from heavily affected people. Nor does fitting a directed model repair confounding or create randomized treatment evidence. The diagnostic "where might there be leverage?" remains distinct from "what effect would this intervention cause?"

Try it: A hospital has the greatest current burden. Does this paper show that resources should be diverted from it? **Answer:** No. A modeled source-control objective is not a complete welfare or triage objective. Such a decision would require intervention evidence, constraints and values that this study does not supply.

D03. What is new relative to connectedness and network control

Where we are: Introduction literature positioning, page 2; References, pages 12–13.

What the passage is doing: The paper connects several existing tools: non-negative vector autoregression, forecast-error variance decomposition, connectedness tables and model predictive control. It does not claim to invent those ingredients. Its proposed contribution is the control-testing layer: use a case-specific symmetrization experiment and a declared prediction record to ask when a directed diagnosis beats reacting to the largest reading. It positions this directed question alongside centrality, immunization and structural-control research.

Worked reasoning: Reason it through by comparing outputs. A descriptive connectedness analysis returns rankings or a matrix. A control experiment returns outcomes under different support rules. A prediction-first record returns a comparison between expectations and later findings. These outputs answer different questions. Merely displaying arrows on a network is not the added control experiment. Conversely, a controller's success without testing why it succeeds would not establish that direction supplied the advantage.

In ordinary language: A new study can use familiar tools to ask a more specific question. Here the question is not simply which node looks important. It is whether the measured direction of influence helps a support rule win a fair model comparison. Credit the tools separately from the proposed test.

Evidence status: INTERPRETIVE positioning supported by cited methods; DEFINITION of the study's verification objects. The draft's broad novelty statements are the author's claims, not a comprehensive independent priority review by this companion.

Do not accidentally conclude: Every directed estimator is new here, or earlier centrality work ignored all useful structure. The paper explicitly compares several established source-aware alternatives and sometimes finds them as good as or better than its named score.

Try it: Which comparison most directly tests the proposed mechanism: a prettier connectedness plot, or repeating the allocation contest while reducing asymmetry? **Answer:** The latter. It changes the hypothesized mechanism within the surrogate while retaining a stated comparison. The plot remains useful for describing what the test is changing.

D04. Transmitter, receiver and loudest unit are different definitions

Where we are: Introduction definitions, page 2; Methods loudest-unit definition, page 9.

What the passage is doing: A transmitter has positive net directional connectedness: its fitted exports exceed its fitted imports. The diagnosed transmitter is the unit with the largest such value. A receiver has the opposite balance. The loudest unit has the greatest realized stress under the domain's declared summary, such as mean or cumulative stress. These definitions do not force the transmitter and loudest unit to be different. Their possible coincidence becomes an important boundary case.

Worked reasoning: Hypothetical arithmetic, not a paper result: a node with exports 6 and imports 2 has net 4 in the stated connectedness units. A node with exports 3 and imports 5 has net −2. Nothing in those two calculations tells you either node's current pollution, incidence or financial loss. You need the stress series for that. Also inspect how “loudest” was summarized: a single peak can select a different unit from a long-run average.

In ordinary language: “Sends most outward” and “has the biggest reading” are different questions. A place can receive much more than it sends. Another can pass a disturbance along while its own reading remains modest. Sometimes the same place leads both lists. Measure the lists before assuming which story applies.

Evidence status: DEFINITION; the arithmetic is a teaching illustration. Fitted transmitter identities later become MEASURED_DATED diagnostic results conditional on the model and data window.

Do not accidentally conclude: A negative net value means a node is harmless, or a positive net value establishes a physical source. Net is a balance of fitted predictive contributions, not a complete causal inventory. A receiver can still transmit some influence and have major needs.

Try it: Could the transmitter be locally quiet and still be a poor real intervention target? **Answer:** Yes. Predictive importance need not be manipulable, and a feasible intervention may not act on the modeled quantity. That additional bridge is not established by the ranking.

D05. The regularity has several distinct null cases

Where we are: Central claim, page 2; mechanism, page 5; falsifiability, page 6.

What the passage is doing: The draft associates an anticipatory advantage with directed asymmetry and a transmitter–loudest gap. It names three ways the advantage can disappear: near symmetry, weak coupling, or a transmitter that is already the loudest unit. These are not interchangeable diagnoses. Near symmetry means little net directional distinction; weak coupling means limited cross-unit influence; coincidence means the two compared rules can choose the same target despite strong direction.

Worked reasoning: Reasoning with rule identity is especially important. If both rules select the same node under the same support schedule, that comparison contains little or no targeting contrast. It does not refute the existence of direction. Influenza illustrates this distinction later. Flight delays illustrate a different limitation: substantial overall connectedness with near symmetry. Armed conflict illustrates weak measured coupling and a problematic pooled domain. A single scalar called “network importance” would hide these differences.

In ordinary language: Two rules cannot gain much from choosing different places when they choose the same place. Nor can a source rule help much when there is little spread to intercept. A system with no stable upstream direction is a third case. Similar small benefits can therefore have different explanations.

Evidence status: INTERPRETIVE regularity evaluated by MODELED_LOCKED experiments and empirical diagnostics. It is not presented as a theorem proved for all non-negative matrices, all budgets or all nonlinear systems.

Do not accidentally conclude: Every asymmetric network must yield a useful real control benefit. The source defines a falsifiable empirical pattern over its tests, while model specification, uncertainty and intervention feasibility still matter. Avoid turning its strong verbal formulation into a universal guarantee.

Try it: A strongly directed network gives no difference because the source is also the loudest node. Is that the same result as a near-symmetric network? **Answer:** No. The observed controller tie may look similar, but one lacks a target contrast while the other lacks the diagnosed directional leverage.

D06. Equation 1: what a first-order VAR says

Where we are: Pipeline, page 3; Methods equation 1, page 9; Figure 1, page 14.

What the passage is doing: Equation 1 is $x_t = \Phi x_{(t-1)} + \varepsilon_t$. The vector x contains all units' stress readings at a time. The matrix Φ maps the previous time's readings into predictions of the next. The residual ε is what that linear prediction leaves unexplained. “First-order” means one modeled lag, not one unit or one cause. Each domain supplies its own stress series: incidence, depreciation, pollution, delay or fatalities.

Worked reasoning: To reason about one matrix entry, hold the other prior readings fixed within the fitted equation. A larger coefficient means more predictive response to that earlier variable in that model's scale. It is not automatically the effect of an intervention that changes the real variable. Units, transformations, omitted drivers and temporal resolution all influence the fitted relation. The model can be useful as a controlled testing substrate while remaining a coarse description of the original system.

Figure 1 should therefore be read left to right. Calibration is not diagnosis, and diagnosis is not observed intervention. The diagram's final box explicitly puts interdiction on a supercritical surrogate. Skipping that label would change the study's claim.

In ordinary language: The model uses the last set of readings to predict the next set. Each predicted reading can depend on several earlier readings. What the rule misses becomes its residual. This gives a common language for different data, but it does not make their underlying biology or physics the same.

Evidence status: DEFINITION of a standard statistical model; MEASURED_DATED fitting; MODELED_LOCKED downstream use. No mechanistic identity across domains is established.

Do not accidentally conclude: A coefficient matrix is a verified pipe, contact or transaction network. This analysis learns predictive dependencies from aggregate series.

Try it: Does “first-order” mean the model captures only direct physical neighbors? **Answer:** No. Any included prior unit may help predict another. The lag restriction concerns time, not verified spatial or physical adjacency.

D07. Non-negativity, ridge and the choice of series

Where we are: Pipeline, page 3; Methods calibration, page 9; Limitations, pages 8–9.

What the passage is doing: The fitted coefficients satisfy $\Phi_{ij} \geq 0$, and ridge regularization stabilizes estimation by penalizing large coefficients. The analysis assesses stationarity per series and uses levels or differences accordingly. These are substantive modeling choices, not merely software settings. A non-negative linear model cannot represent an inhibitory lag in the same way as an unconstrained model. Differencing changes the target from a level to a change.

Worked reasoning: Consider the short flight-delay series mentioned in the limitations: roughly 90 time points for 18 units. Many possible cross-unit coefficients must be estimated from limited temporal information. Shrinkage reduces instability, but it does not manufacture new observations. A ranking can remain fragile even after a fitting routine converges. Dependence over time also means that 90 rows are not 90 independent experimental replications.

The currency example supplies another warning. Its use of non-stationary levels is explicitly retained as a limitation. A companion should not silently repair that choice or describe all cases as equally well supported. The common pipeline provides comparability of procedure, not identical statistical difficulty.

In ordinary language: Restrictions can keep a model from chasing noise. They also limit what it can describe. A rule that permits only positive fitted links is built to study reinforcing spread. It may miss a real balancing response. Always ask which data scale and constraints produced a ranking.

Evidence status: DEFINITION and SCOPE of estimation; empirical diagnostics are conditional on these choices. The draft refers detailed stationarity results to its supplementary material rather than printing them all here.

Do not accidentally conclude: Constraints remove all overfitting, prove the arrows causal or guarantee stable rankings. A short series and a persistent trend can still influence the result.

Try it: Would a transmitter ranking on levels necessarily equal one on changes? **Answer:** No. They ask about different variation. Record the transformation and test sensitivity instead of treating the two targets as interchangeable.

D08. From residuals to a connectedness table

Where we are: Pipeline, page 3; Methods connectedness, page 9; Figures 1–2, page 14.

What the passage is doing: Generalized forecast error variance decomposition, or GFEVD, divides model forecast-error variance into contributions associated with different units' innovations. The fitted coefficient dynamics and residual covariance both enter this construction. The Diebold–Yılmaz table summarizes those directed contributions as “to,” “from” and net connectedness. Total connectedness aggregates sharing, while net connectedness compares outward and inward contributions for a particular unit.

Worked reasoning: A useful reading exercise is to cover the labels on a connectedness chart and identify its axis. A positive net bar is not a pollution concentration, a number of infections or a percentage reduction under support. Figure 2's middle panel uses net connectedness percentages; its left panel uses measured PM2.5; its right panel uses surrogate reductions. All three belong in the explanation, but they cannot be added or compared as if they had the same denominator.

Residual covariance matters because simultaneous unexplained movement can affect the decomposition. “Generalized” does not mean immune to common drivers, omitted variables or model error. The paper itself keeps predictive directional influence separate from intervention effects and asks for further causal-invariance checks in volatility settings.

In ordinary language: The fitted model asks which units' surprises help account for its forecast uncertainty. Adding those contributions gives an overall sharing measure. Comparing outward and inward contributions gives a role for each unit. The same overall total can hide very different balances among individual units.

Evidence status: DEFINITION of the diagnostic; MEASURED_DATED estimates from fitted series. The control implications require the subsequent MODELED_LOCKED tests.

Do not accidentally conclude: “Connectedness” is not a direct count of physical links. In particular, the largest total sharing does not identify the best support target.

Try it: Can two networks have similar total connectedness but opposite net-role patterns? **Answer:** Yes. An aggregate can stay similar while the directional distribution changes. The smoke/flight comparison and symmetrization test investigate precisely that distinction.

D09. Equation 2: horizon integration and its evidence boundary

Where we are: Methods equation 2 and horizon statement, page 9; Limitations horizon sensitivity, page 8.

What the passage is doing: Forecast-variance contributions depend on how far ahead the model looks. Equation 2 writes an integral of net_i(H) over forecast horizon H. The draft states that maximizing this integrated net score gives the same transmitter identity as its fixed-horizon analysis in the networks it recomputed. It also reports stability over multi-step horizons $H \geq 4$. The PDF does not provide a general proof that every possible network preserves its rank under every horizon range.

Worked reasoning: Reason it through without inventing measurements. One unit could have a strong immediate effect that fades; another could act through a longer chain. Their accumulated contributions need not rank the same at all horizons. Integration also needs a stated range and weighting to define a numerical score. When applying the idea elsewhere, those details belong in the analysis plan rather than being hidden by the integral symbol.

The source's phrase about eliminating a free horizon parameter should therefore be taught alongside its actual reported verification. The printed evidence is identity agreement on studied networks and referenced supplementary sensitivity analysis. No new horizon calculations were performed for this guide.

In ordinary language: A short look ahead and a long look ahead can tell different stories. The paper checks whether its leading source changes when it changes that view. In its reported cases, the selected source stays the same. Treat that as a useful check of those cases, not a promise about every new system.

Evidence status: DEFINITION of an integrated summary; MEASURED_DATED robustness claim for evaluated cases; SCOPE for generalization. The notation alone is not a universal rank-invariance theorem.

Do not accidentally conclude: A horizon choice can never matter, or that an unspecified integral automatically removes every tuning decision.

Try it: A new system changes its leading transmitter between short and long horizons. Should you suppress that result because equation 2 appears in the paper? **Answer:** No. Report the instability, specify the horizon target and investigate whether a single static source summary is appropriate.

Making the allocation experiment interpretable

D10. “Loudest” depends on the declared observation summary

Where we are: Introduction, page 2; Methods loudest definition, page 9; smoke traces, Figure 2a, page 14.

What the passage is doing: The paper operationalizes loudest as the greatest realized stress, summarized by cumulative or mean series depending on the domain. It specifically warns that a peak can select a different state from a mean in the smoke data. This choice matters because the reactive comparator is defined through that target. Changing the summary changes the rule being compared, even if the fitted network is left untouched.

Worked reasoning: Think through the Figure 2 traces. Tall isolated spikes and sustained elevations convey different features of a pollution episode. A maximum emphasizes the worst instant. An average emphasizes the typical level over the chosen window. A cumulative summary depends on the time span and observation spacing. None automatically measures population harm, because exposure, population and vulnerability are not supplied by the stress ranking alone.

This also clarifies “reactive.” The paper's comparator is a particular damage-weighted heuristic with a fixed support budget. It is not a trial of every emergency-response procedure that responds to damage. A strong comparison names the implemented rule rather than treating its label as an entire field of practice.

In ordinary language: The place with the sharpest bad day need not have the largest burden over the whole period. Decide which meaning you intend before ranking places. Then use that meaning consistently. A ranking is not wrong merely because it answers a different question, but it must be named honestly.

Evidence status: DEFINITION and SCOPE. The domain-specific selections are reported analysis choices; their ethical or operational superiority is not evaluated.

Do not accidentally conclude: The labeled loudest node has the greatest population exposure or deserves the most assistance under all objectives. Those are different measurements and decisions.

Try it: If you replace the mean with a peak and get another “loudest” state, have you disproved the transmitter calculation? **Answer:** No. You changed the reactive target definition. Report both definitions and compare the resulting allocation tests under the same conditions.

D11. Why rescale the fitted matrix to a common growth regime?

Where we are: Pipeline, page 3; Limitations, pages 8–9; Methods interdiction, page 10.

What the passage is doing: Native fitted models differ in stability. Some are stable; the smoke fit is reported near spectral radius 1.28. The controller experiments rescale each matrix to a mildly supercritical radius of 1.05 or 1.06. The spectral radius is the largest absolute eigenvalue. In this linear setting it helps describe whether repeated propagation tends to decay or amplify. The rescaling makes the cascade contests more comparable, while changing their distance from the native empirical dynamics.

Worked reasoning: A worked interpretation uses the same matrix in two roles. The fitted coefficients describe the calibrated surrogate before rescaling. A scalar multiplication then adjusts its growth tendency for the control contest. This preserves some structural relationships while changing time evolution and absolute cascade magnitudes. It is therefore reasonable to ask about a structural pattern across the contests, but unreasonable to report a percentage reduction as if the original system had undergone that support intervention.

The draft says the qualitative pattern survives its radius sensitivity checks. Those supplementary checks are not re-created here, and their claim does not make the absolute results invariant. Ratios such as “2.5 times” belong to a specified model regime too.

In ordinary language: A controller has little to stop if the model settles down by itself. The paper sets up a common, mildly growing challenge so the rules face a comparable test. That is a deliberate experiment on a model. It is not a finding that every real network was growing at that rate.

Evidence status: DEFINITION of spectral radius; MODELED_LOCKED rescaling experiment; SCOPE for effect sizes.

Do not accidentally conclude: A near-unit fitted radius certifies a physical phase transition or measures a real disaster threshold. Persistence, trends and estimation can also produce a large fitted radius.

Try it: Can you use the 42% equity reduction as an estimated bailout effect after learning about rescaling? **Answer:** No. It is an outcome of the rescaled surrogate, not an identified effect of a historical or proposed bailout.

D12. A fair controller contest needs a shared budget and baseline

Where we are: Pipeline, page 3; Benchmark, page 4; Methods interdiction, page 10.

What the passage is doing: The paper compares no action, the loudest-node heuristic, undirected correlation centrality, VAR out-strength, impulse-response spillover, static transmitter targeting and dynamic cross-entropy model predictive control. Each intervention allocates a fixed per-step budget in the surrogate. Cascade reduction is defined relative to the no-action model outcome, not to a clinical or financial utility function. A negative reduction means the simulated outcome is worse than no action.

Worked reasoning: A hypothetical interpretation, not a new simulation: if a controller increases the modeled cascade, its reduction is negative even if it gives useful local relief to one node. That tension reflects the chosen aggregate objective. It does not prove local assistance is inherently harmful. Objectives can disagree, and the study evaluates only its specified cascade objective.

MPC repeatedly searches possible support plans using its model. “Cross-entropy” names the stochastic search approach; it does not make the forecasts true or the plan globally optimal in every run. Static targeting is simpler because it computes a role and keeps using it. Comparing these rules asks whether added planning machinery improves the particular surrogate outcome beyond a good fixed diagnosis.

In ordinary language: To compare rules, give them the same resources and the same modeled problem. Then state what counts as improvement. A rule that receives more support or sees future information is answering a different contest. The no-action result is the reference point for the reported reductions.

Evidence status: DEFINITION of comparison rules and MODELED_LOCKED outcomes. Detailed implementation belongs to the deposited analysis, not an invented optimization recipe in this guide.

Do not accidentally conclude: No-action is an acceptable real emergency policy, or the simulation objective captures all harms and benefits of support. It is an experimental comparator.

Try it: Why keep budget and random seeds shared across controllers? **Answer:** They reduce avoidable differences in resources and simulated conditions, making the remaining outcome contrast more informative about the rule. They do not resolve model misspecification.

D13. Equation 3 removes the antisymmetric matrix component

Where we are: “Isolating the asymmetry,” page 3; Methods equation 3, page 10.

What the passage is doing: Equation 3 is $\Phi(\alpha) = \alpha\Phi + (1-\alpha)(\Phi+\Phi^{T})/2$, evaluated at $\alpha = 1, 0.75, 0.5, 0.25$ and 0. Let $S = (\Phi+\Phi^{T})/2$ and $A = (\Phi-\Phi^{T})/2$. Then $\Phi = S+A$, and substitution gives $\Phi(\alpha) = S+\alpha A$. Before the later rescaling, S is unchanged while the antisymmetric part is multiplied by α. This algebra identifies exactly what the interpolation changes.

Worked reasoning: A symbolic worked example avoids false empirical precision. For two opposite matrix entries a and b, the symmetric contribution is their average. Their directional difference is scaled by α. At $\alpha = 1$

the original difference remains; at $\alpha = 0$ it vanishes. This statement is about matrix entries. It does not by itself prove that every forecast-variance statistic or controller outcome changes linearly with α.

The controller contest must be recomputed along the path. Simply drawing a symmetric diagram would not test the outcome mechanism. Nor should a smooth-looking curve be treated as independent evidence from each point: all points are transformations of the same fitted case.

In ordinary language: Pair each fitted direction with its reverse and average them. That makes a symmetric version of the matrix. The test moves from the original toward that version in stated steps. It asks whether the source-targeting edge fades as the directional imbalance is removed inside the model.

Evidence status: ANALYTIC for $S+\alpha A$; MODELED_LOCKED for the controller response along the path. The latter depends on the simulation and rescaling.

Do not accidentally conclude: The operation experimentally removes real-world causal direction, or preserves every connectedness total. In particular, post-interpolation scalar rescaling can change the symmetric component's numerical magnitude.

Try it: What remains fixed in the algebra before rescaling? **Answer:** S, the symmetric matrix component. What is attenuated? A, the antisymmetric component. The separate radius adjustment must be included when describing the full simulation procedure.

D14. What the symmetrization null controls, and what it does not

Where we are: Symmetrization narrative, page 3; equation 3 and radius adjustment, page 10; uncertainty, pages 8–9.

What the passage is doing: After each interpolation, the paper re-rescales the matrix to the chosen common spectral radius. Otherwise changing direction could also change the system's tendency to grow, making the controller comparison harder to interpret. The null is therefore a controlled model perturbation: compare outcomes as asymmetry changes at a fixed specified growth setting. The source reports that advantages collapse in confirmatory cases and remain absent in baseline cases.

Worked reasoning: Reason through the two operations in order. First, the unscaled blend retains S and reduces A. Second, a scalar multiplication fixes the radius. A precise explanation cannot claim that the final matrices have identical symmetric entries, identical residual effects, identical total connectedness and identical control landscapes. The controlled feature is the chosen radius; other derived quantities may change as the model is recomputed.

The draft also refers to phase-randomisation nulls in its discussion. Those provide another referenced robustness object, but the selected PDF does not spell out their full procedure or results. This guide records that role without supplying unprinted numbers. Reading a source completely includes recognizing when it delegates detail elsewhere.

In ordinary language: Changing one part of a model can accidentally change another. The paper resets the growth setting after reducing direction. That makes a loss of advantage harder to explain simply by an easier, shrinking cascade. It still remains a test within the fitted model.

Evidence status: MODELED_LOCKED mechanism check; ANALYTIC description of the operations; SCOPE regarding unprinted supplementary details.

Do not accidentally conclude: A successful symmetrization test proves all alternative explanations impossible or establishes a universal theorem. It isolates a specific aspect of a particular surrogate experiment.

Try it: Why is “same growth regime” more accurate than “everything except direction is identical”? **Answer:** The procedure explicitly fixes the radius, but interpolation and rescaling can affect other derived statistics. The narrower description states the actual control rather than promising unsupported invariance.

Reading each empirical comparison

D15. Figure 2 and the near-matched smoke–flight comparison

Where we are: Connectedness comparison, pages 3–4; Figure 2a–c and caption, page 14; Figure 4a, page 15.

What the passage is doing: Smoke has about 82% total connectedness; flight delays about 77%. This is a near match, not numerical equality. The smoke model has a Wisconsin transmitter and Michigan mean-loudest receiver. Its support contest gives roughly 10% reduction for transmitter targeting versus 4% for the loudest rule. Flight delays are near symmetric and controllers remain near baseline. The comparison illustrates why a high total sharing score alone does not predict an advantage.

Worked reasoning: Read all three Figure 2 panels separately. Panel a displays dated pollution traces in concentration units. Panel b displays fitted net connectedness. Panel c compares surrogate cascade reductions. Its other displayed methods also show that the named transmitter score is not the only competitive source-aware option. The caption attaches the important uncertainty qualifier: these are seed-averaged point estimates, and smoke's block-bootstrap advantage interval includes zero.

The approximate ratio 10/4 explains the stated 2.5-fold comparison. It is a ratio of modeled reductions in this regime, not a measured reduction in pollution or a portable multiplier for air-quality response. A small denominator can also make a ratio sound more dramatic than the absolute gap.

In ordinary language: Two systems can be strongly linked in total while organizing that link very differently. Smoke supplies a fitted upstream–downstream contrast. The flight example does not supply the same useful contrast. Look at the role pattern and model test, not just the total percentage.

Evidence status: MEASURED_DATED traces and fitted diagnostics; MODELED_LOCKED controller results. The source itself calls the pair structurally illustrative rather than individually decisive.

Do not accidentally conclude: The pair proves all high-connectedness networks behave differently only because of direction. It is one comparison, strengthened by a within-case perturbation and qualified by uncertainty.

Try it: Which phrase is faithful: “identical connectedness” or “near-matched total connectedness”? **Answer:** Near-matched. Retaining that distinction avoids overstating the experimental control supplied by the pair itself.

D16. The equity anchor: large surrogate contrast, narrow interpretation

Where we are: “The clearest case,” page 4; Limitations, pages 8–9.

What the passage is doing: The equity case aggregates eight national indices weekly over 2007–2010. Total connectedness is reported as 81%; the United States has dominant net connectedness around +21%, while Hong Kong and Japan are receivers. The model support comparison is about +42% reduction for transmitter

targeting and −7% for the reactive rule. The approximate advantage of +49 percentage points becomes −23 under the symmetrized experiment. Equity is separate from the six out-of-domain tests but included in the seven-network origin-recovery audit.

Worked reasoning: A numerical reading check distinguishes reduction from advantage. The 42% and −7% are each compared with no action. Their difference is roughly 49 percentage points. The subsequent +49 to −23 statement refers to that advantage under a model transformation. It should not be retold as an observed market return changing from a gain to a loss.

The historical origin matters to the recovery audit, but knowing a familiar crisis narrative can also shape expectations. The declared source hypothesis and null selection rule are therefore part of the evidence. The paper additionally names volatility-spillover concerns as a future causal-invariance check rather than treating the financial arrows as settled mechanisms.

In ordinary language: In this fitted financial simulation, the two rules have a large gap. The reactive rule even worsens the modeled aggregate outcome. The test becomes more informative when that edge reverses as the matrix is made symmetric. None of these numbers is a measured result of a bailout or a trading strategy.

Evidence status: MEASURED_DATED calibration and historical-origin comparison; MODELED_LOCKED effects. Equity and held-out COVID-19 are the two cases with individual advantage intervals excluding zero in the reported bootstrap.

Do not accidentally conclude: The paper establishes how financial authorities should distribute support. Real institutions, feasible instruments, welfare objectives and intervention identification are not modeled adequately for that claim.

Try it: Is a −7% reduction a 7% improvement? **Answer:** No. Under the stated reduction convention it means the simulated cascade is larger than no action. Keep the sign and comparator explicit.

D17. Figure 3 compares estimators, not just two slogans

Where we are: Benchmark, page 4; Figure 3a–c and caption, page 15.

What the passage is doing: Figure 3 includes a ground-truth recovery panel, a synthetic control panel and a held-out COVID-19 control panel. Directed influence measures include VAR out-strength, impulse-response spillover and Diebold–Yılmaz net connectedness. Their empirical-surrogate reductions span about 35–65%, against about 5% for correlation centrality, 0.2% for reactive triage and 30% for MPC. The plotted reactive bar rounds to zero, while the caption gives 0.2%; both describe a negligible result at different precision.

Worked reasoning: Read panel a as rank agreement with a specified support-target truth, not as the fraction of real interventions that succeed. The reported directed correlations are roughly 0.5–0.8; the undirected correlation is around −0.1. Panels b and c instead report reductions in separate surrogate experiments. Their bars have different magnitudes and should not be pooled into one estimated average benefit.

The synthetic twin offers a controlled setting with known structure. The empirical twin calibrates a model from observed series. The word “real” in the panel label refers to that calibration source; the control remains simulated. This is exactly where a reader can accidentally promote a figure caption into field evidence.

In ordinary language: Ask whether the proposed idea survives strong alternatives. Here several ways of finding source structure work well. The result is broader than “one special score wins.” A costly planner also has to beat simple source-aware rules, not merely a weak reactive rule.

Evidence status: MODELED_LOCKED benchmarks, with MEASURED_DATED empirical calibration. The comparison supports a structural pattern in the tested experiments, not uniqueness of one estimator.

Do not accidentally conclude: MPC is generally inferior, or a 65% bar predicts epidemic prevention. Computational effort, objective, model and protocol determine the comparison.

Try it: Why retain the caption's 0.2% when the bar says zero? **Answer:** It prevents inventing an exact null from a rounded graphic. Explain that the result is small rather than claiming the source reported literally no change.

D18. Cross-paradigm alternatives keep the baseline challenge honest

Where we are: Benchmark closing paragraph, page 4; cited control literature, pages 12–13.

What the passage is doing: Beyond within-pipeline scores, the draft reports comparisons with effective-distance source centrality, collective influence and structural-controllability driver nodes on the two anchor networks. Transmitter targeting matches collective influence on equity and trails it modestly on held-out COVID-19. The point is that methods using source structure outperform the reactive comparator, not that the named connectedness score is unbeatable. Detailed numerical comparisons are delegated to supplementary information.

Worked reasoning: A conceptual comparison helps. Out-strength summarizes fitted outward coefficients. Effective-distance methods use a notion of reach through a network. Collective influence and driver-node approaches come from other intervention or control formulations. These are not automatically interchangeable, and their formal objectives can differ. The manuscript's empirical comparison places them into its common support contest; the guide should not invent full derivations or claim their usual theoretical guarantees transfer unchanged.

When only a qualitative result is printed, teach that result at its printed precision. “Competitive,” “matches” and “trails modestly” are supported descriptions here. An exact unprinted percentage or an overall winner across all domains would be an added empirical claim. The absence of those details in the selected PDF is a boundary of this companion's coverage, not a reason to manufacture them.

In ordinary language: A new method should face alternatives built from different ideas. If a rival also succeeds, that can support the wider structural explanation while limiting claims about which tool is best. Say both parts. Do not hide a strong rival because the main figure favors your chosen method.

Evidence status: Reported MODELED_LOCKED supplementary comparison; INTERPRETIVE conclusion about source structure. The guide has not independently rerun the alternatives.

Do not accidentally conclude: Every source-aware method is equally good, or success of any one proves causal source identification. The shared evidence remains model-based.

Try it: Does a rival beating transmitter targeting on one anchor refute every directedness claim? **Answer:** No. It limits estimator-specific superiority. The broader claim concerns whether useful source structure beats the reactive rule under the declared experiment.

D19. The upstream explanation is a mechanism story within the model

Where we are: “Why directedness governs control,” page 5; predictive-causal distinction, pages 2 and 9–10.

What the passage is doing: The paper explains its pattern through upstream interception. Supporting a transmitter can reduce propagation before downstream units accumulate stress; supporting a sink can leave the incoming propagation intact. The explanation is intuitive in the fitted reinforcing model. It also explains why a symmetric network, weak coupling or source–loudest coincidence can remove the comparison's advantage. It is not an independent physical experiment confirming the arrows.

Worked reasoning: Reason through a limitation of the metaphor. A real source may be difficult to alter, or the action may change behavior in a way the linear surrogate omits. A downstream intervention might prevent harms not counted by the cascade objective. It could also affect feedback paths that the simple upstream–sink story suppresses. Those possibilities do not invalidate the model experiment; they limit translation from a favorable model comparison to a response recommendation.

The draft's actual mechanism test is stronger than the metaphor because it changes asymmetry and reruns the support contest. Even that test has a model-specific ceiling. The supporting interpretation should be stated as "consistent with interception of modeled directed spread," rather than "proves how the real cascade was caused."

In ordinary language: Stopping a leak upstream can be more effective than mopping downstream. That picture makes the proposed allocation rule easy to understand. But the fitted model must still show that there is an upstream relation to exploit. A useful metaphor cannot supply missing causal evidence.

Evidence status: INTERPRETIVE explanation of MODELED_LOCKED findings; not new MEASURED_DATED intervention evidence.

Do not accidentally conclude: A quiet node must be the source, or every loud node is merely a passive sink. Those are empirical/model diagnostic questions, not definitions of quietness and loudness.

Try it: Can the leak-and-mop analogy identify the correct node without data? **Answer:** No. It explains why a source could matter after a valid diagnosis. It cannot establish which node has that role or whether the available intervention changes its exported influence.

D20. What the prediction-first protocol commits to

Where we are: Six-test introduction, page 5; Methods protocol, page 10; availability qualification, page 11; Figure 4, page 15.

What the passage is doing: Before network computation, the author reports writing a hypothesized transmitter, the expected transmitter–loudest gap, unit-selection rule, estimator and falsification criteria. The original five records lack an externally verified public pre-analysis timestamp, so the paper explicitly calls them prediction-first rather than formally preregistered. The held-out COVID-19 test is counted separately. Together they yield three confirmations, one refinement and two failed predictions; equity is an additional network, not an extra member of the six.

Worked reasoning: A bookkeeping exercise prevents several errors. There are five initial prediction-first domains, one held-out COVID-19 comparison and one separate equity anchor. The five domains yield two confirmations, one refinement and two failures. Adding held-out COVID-19 gives the six-test 3/1/2 tally. Adding equity gives seven networks for the origin-recovery audit, a different denominator and question.

The three additional OSF predictions concern systems not analyzed in this draft. Their external registration demonstrates a prospective extension of the protocol, not retroactive preregistration of the initial five or extra successful results. Keep existence, timestamp, analysis and outcome as distinct facts.

In ordinary language: Write down what you expect before looking at the result, including what would count against you. Then keep the misses visible. A later public archive shows that the record existed by its deposit date. It cannot by itself prove when each expectation was first written.

Evidence status: SCOPE and MEASURED_DATED recordkeeping claims, with an explicit timestamp limitation. The companion has not independently reconstructed private drafting chronology.

Do not accidentally conclude: The archive date verifies every pre-analysis commitment, or that the three later registrations increase the published confirmation count.

Try it: Why is “seven preregistered successes” wrong? **Answer:** It combines different denominators, erases two failed expectations and upgrades the initial records beyond their verified timestamp status. The correct labels are part of the evidence, not optional modest wording.

D21. Asian currencies: a confirmation with a persistence caveat

Where we are: Confirmations, page 5; Limitations, page 9; Figure 4, page 15.

What the passage is doing: For nine currencies in the 1997 Asian crisis, the prediction identifies the Thai baht as a quiet historical origin and the Korean won as the louder receiver. The fitted network matches that expectation. The surrogate comparison gives approximately +13% versus +6% reductions, summarized as about 2.3-fold in the draft. The rounded percentages and rounded ratio should not be treated as an exact arithmetic identity. This network is also estimated on non-stationary levels, a disclosed limitation.

Worked reasoning: Reason through the evidence chain. First comes a historical hypothesis. Then comes a fitted net-role calculation. Then comes a control comparison in a rescaled surrogate. Agreement at the first two stages is not a measured benefit at the third stage in the real currency system. Nor does a familiar historical story make the estimated direction uniquely causal.

Figure 4 supplies a total connectedness value around 54% and classifies the domain as a confirmation. Its caption warns that each displayed twin's individual advantage interval includes zero. The origin match and the within-case perturbation therefore carry a different kind of weight from a precise intervention magnitude. Teaching the successful classification while omitting that uncertainty would overstate the case.

In ordinary language: The historical source and the worst-looking series need not be the same. This case supports the predicted distinction in the fitted network. Yet a trend or persistent level can affect a time-series model. Keep that modeling concern next to the successful source match.

Evidence status: MEASURED_DATED fitted recovery; MODELED_LOCKED reductions; SCOPE for non-stationary levels and uncertainty.

Do not accidentally conclude: A familiar crisis origin validates all fitted coefficients, or a near-unit spectral radius certifies a physical critical transition. Persistence is an alternative explanation for that numerical feature.

Try it: Should the currency result be deleted because it has a limitation? **Answer:** No. Report the result and the limitation together. A transparent boundary is more useful than either unqualified success or silent exclusion.

D22. Smoke: a successful prior classification is still not field control

Where we are: Confirmations, page 5; Figure 2 and caption, page 14; Figure 4, page 15.

What the passage is doing: The smoke case predicts a distinction between an upwind transmitter and a downwind loudest receiver. The fitted diagnosis identifies Wisconsin and Michigan respectively. Seventeen states provide the analysis units; the figure also shows contextual city/county traces, which should not be mistaken for additional modeled states. The paper reports a 2.5-fold surrogate comparison, around 10% versus 4%, and checks that the advantage disappears with symmetrization.

Worked reasoning: There are several possible meanings of “source” here. The physical fire origin, a fitted transmitting state in aggregate pollution series and a feasible point of intervention are not the same object. The source draft labels Wisconsin as an upwind transmitter within its analyzed state network. A faithful companion must not silently turn that into the ignition location of the fires or a measured transport model.

Read the caption's uncertainty with the confirmation label. “Confirmation” describes agreement with a declared structural prediction. It does not mean the smoke controller's individual advantage is precisely estimated; its block-bootstrap interval spans zero. This distinction lets a reader understand why the paper includes the case while assigning more individual statistical weight to equity and COVID-19.

In ordinary language: A state with high pollution can be receiving the episode rather than leading the fitted spread. The model result supports looking for that difference. It does not tell an air-quality authority that a particular real action in Wisconsin would reduce Michigan's pollution by the plotted amount.

Evidence status: MEASURED_DATED pollution and diagnostic role; MODELED_LOCKED support contest; INTERPRETIVE transport story within the aggregate analysis.

Do not accidentally conclude: A predictive state role establishes verified atmospheric transport edges or a validated emissions-control policy.

Try it: Is Wisconsin being called the physical location where the fires began? **Answer:** No. It is the fitted transmitter among the selected state series. Changing the modeled unit system or the intervention question could change what source means.

D23. Influenza refines the target contrast

Where we are: Refinement, page 5; Figure 4, page 15.

What the passage is doing: The influenza analysis covers fourteen states over 2010–2019. It has strong directedness, with total connectedness around 78%, but Texas is both the fitted transmitter and the loudest state. Transmitter targeting therefore improves only marginally over the loudest rule, about 1.1-fold in the reported comparison. The paper treats this as a refinement: direction alone is not enough to create an allocation advantage if the compared rules select the same place.

Worked reasoning: Reason through what is being refined. The broad slogan “asymmetry helps” leaves out the decision contrast. A source-aware rule can differ usefully from a damage-based rule only if their targets or schedules differ in a consequential way. Influenza adds that operational condition. The refinement is scientifically useful because it narrows a claim that could otherwise be made too broadly.

The plotted values remain surrogate outcomes, with wide individual uncertainty. The case does not show that influenza intervention resources should always go to the state with greatest reported burden, or that Texas is a fixed biological origin of seasonal influenza. The fitted role is conditional on selected states, aggregation and time span. A different epidemic wave or sampling system could alter it.

In ordinary language: A good diagnosis may agree with the simple rule. Then there is little benefit from using one label instead of the other. That does not mean the network lacks direction. It means the practical difference between the choices is small in this setup.

Evidence status: MEASURED_DATED role coincidence and MODELED_LOCKED comparison; INTERPRETIVE refinement of the regularity.

Do not accidentally conclude: A small controller gap falsifies every role diagnosis, or that source-aware and reactive policies are universally equivalent in influenza. The reported coincidence is case-specific.

Try it: What extra question should be added after finding strong directedness? **Answer:** “Does the proposed source rule choose something meaningfully different from the reactive comparator under the actual support protocol?” That question turns a structural pattern into an honest comparison rather than assuming an advantage from a label.

D24. Flight delays: preserve the failed expectation

Where we are: Falsifications, pages 5–6; connectedness pair, pages 3–4; Figure 4, page 15.

What the passage is doing: For eighteen airports during the 2013–2014 polar-vortex winter, the author expected a directed front from a weather-exposed hub. Instead, the fitted network is highly connected, around 77%, but near symmetric. A warm Sun Belt airport emerges as the net transmitter; heavily disrupted weather hubs are receivers. Controllers cluster near the no-action baseline. This fails the prior domain prediction even though the fitted near-symmetric structure is consistent with the regularity's null region.

Worked reasoning: The reasoning distinction is between prediction and conditional interpretation. The prior statement assigns a domain to a structural category. The analysis then tests that assignment. Separately, the controller result can be checked against the measured category. Success at this second conditional check does not erase failure at the first. The paper's confirm–refine–falsify record is informative only if that bookkeeping remains visible.

This is also a relatively short series, around 90 time points for eighteen units. Common weather shocks, temporal aggregation and limited sampling can affect fitted roles. A near-symmetric aggregate model is not a statement that aircraft, passengers and delays never move directionally through real routes. The diagnosis concerns the chosen series and model.

In ordinary language: The story expected before analysis was wrong. The later model result can still help explain why the controller had little advantage. Keep both facts. A useful after-the-fact explanation does not turn an incorrect forecast into a correct one.

Evidence status: MEASURED_DATED failed structural expectation; MODELED_LOCKED small advantages; SCOPE for interpretation.

Do not accidentally conclude: The method predicted the flight null in advance, or that the analysis proves airport interventions cannot work. Neither follows from this particular failed hypothesis and surrogate contest.

Try it: Can you write “flight delays confirmed the original prediction because no controller won”? **Answer:** No. The original directed-front prediction failed. The later null interpretation is a different, conditional statement and must be described that way.

D25. Armed conflict: a confounded null has limited reach

Where we are: Falsifications and scope, page 6; Figure 4, page 15; data availability, page 11.

What the passage is doing: The conflict analysis uses six countries over 2012–2023 from UCDP's Georeferenced Event Dataset. The prior hypothesis expects the Malian origin to transmit. The fitted model instead assigns it a receiving role and shows weak coupling, around 21% connectedness, with controllers near baseline. The source explicitly calls this a confounded null because it pools two largely decoupled insurgencies at coarse monthly resolution.

Worked reasoning: Reason through aggregation. If selected units do not belong to one strongly coupled process, a single common model can average over different dynamics. Coarse time bins can also miss sequence. A source that matters at another spatial or temporal scale may not appear as the dominant net role in these monthly country totals. That possibility is a methodological caveat, not a license to invent the desired answer.

The ethical stakes are another reason to keep the interpretation narrow. A fitted role in aggregated fatalities is not responsibility, intent, culpability or a recommendation for real coercive action. The study's simulated support variable does not represent the full consequences of conflict policy. A response system would need verified domain mechanisms and specialist judgment beyond this model.

In ordinary language: Combining distinct processes can hide the pattern you hoped to study. A weak combined signal may reflect the way the data were grouped. This case records a failed expectation and a boundary of this analysis. It is not evidence that real conflict has no spreading mechanisms.

Evidence status: MEASURED_DATED diagnostic null and failed origin expectation; MODELED_LOCKED control tie; SCOPE for pooling and resolution.

Do not accidentally conclude: Mali's fitted receiving role disproves historical accounts, or that the common surrogate validates a conflict intervention. Both would promote an aggregate predictive analysis into an unsupported causal claim.

Try it: What should follow a weak result from pooled, partly separate processes? **Answer:** Report the null and its construction, then consider a separately specified analysis with appropriate units and timing. Do not relabel the original prediction as successful after changing the question.

D26. What would falsify the structural claim?

Where we are: Falsifiability, page 6; Discussion testing record, pages 7–8; Figure 4 caption, page 15.

What the passage is doing: The draft names two anomalies that would challenge its regularity: a directionally asymmetric network where transmitter targeting loses, or a symmetric network where it wins with an advantage surviving the symmetry test. Neither is reported. It also distinguishes those structural checks from failed prior domain classifications. Because classification partly uses the same fitted model and diagnostic experiment, the tests should be described precisely rather than treated as unrestricted independent proof.

Worked reasoning: Reason through a genuinely challenging new case. Suppose a source-aware rule repeatedly loses despite a robust source–receiver distinction, under matched budgets and a well-specified model. That would directly challenge the proposed operational link. Before declaring a discovery, one would still inspect implementation, uncertainty and whether “directional” was measured independently of the performance label. A falsification criterion is useful only if it can be applied without silently redefining success.

The source's broad wording about “precisely when” is stronger than a finite set of case studies can prove universally. This companion therefore teaches the stated regularity and its announced vulnerability, while retaining the explicit limitation that no general theorem is supplied. Absence of an anomaly in these tests is evidence within their scope, not proof that one cannot exist.

In ordinary language: Say what result would make you change your mind. Then separate that from guessing the wrong category before seeing data. The paper's two domain misses are real misses. The narrower structural story can still fit the later results, but its support must be judged on that narrower question.

Evidence status: INTERPRETIVE formulation of testable scope; MODELED_LOCKED and MEASURED_DATED test outcomes.

Do not accidentally conclude: Failure to find a counterexample establishes a universal law, or that every failed domain forecast is merely a success under another name.

Try it: Why should a new asymmetric losing case remain in the report? **Answer:** It addresses the central structural prediction directly. Excluding it because it disagrees would destroy the test's stated vulnerability and make the claimed regularity less informative.

Static diagnosis, changing networks and uncertainty

D27. Figure 5 asks how much planning adds beyond a fixed source rule

Where we are: “On stationary surrogates,” pages 6–7; Figure 5a–b and caption, page 16.

What the passage is doing: The paper decomposes an anticipatory controller's advantage into fixed directed reallocation and repeated planning. In a synthetic near-stationary network, static targeting captures about 96% of MPC's reduction. In the held-out COVID-19 decomposition, static targeting gives about +32% versus +28% for MPC. These magnitudes use a protocol distinct from Figure 3. Their within-experiment ranking, rather than a pooled comparison across figures, carries the intended lesson.

Worked reasoning: Read the ratio carefully. “96% of the planner's reduction” is not a 96-percentage-point improvement over no action. It compares two reductions in the same experiment. Likewise, the empirical panel's static rule slightly exceeding MPC does not mean the exact same values should appear in Figure 3. The source explicitly distinguishes those protocols. Different support settings and decomposition questions can produce different magnitudes without contradiction.

The synthetic setting supplies a controlled example of a stable useful source. The empirical calibration shows a similar ordering in one decomposition. Neither tests every nonlinear process, information delay or operational constraint. “Static suffices” therefore belongs to the near-stationary, first-order surrogate setting studied here.

In ordinary language: Before building a complicated planner, test a good simple rule. If the important source stays put, choosing it once may capture most of the gain. That makes diagnosis valuable even when repeated optimization adds little. It does not show that every fixed rule works in changing systems.

Evidence status: MODELED_LOCKED decomposition experiment. “Stationary” here concerns the stability relevant to the targeting comparison, not a claim that every observed series is constant.

Do not accidentally conclude: The 32% bar replaces the 35–65% benchmark, or a static diagnosis will remain useful indefinitely without monitoring.

Try it: A proposed MPC beats reactive triage. What additional baseline should be requested? **Answer:** A strong fixed source-aware rule under the same model, budget, information and protocol. That comparison tests whether repeated planning adds value beyond the diagnosis itself.

D28. Figure 6: a moving role makes updating worth testing

Where we are: Non-stationary systems, page 7; Figure 6a–b, page 16.

What the passage is doing: Rolling windows refit the COVID-19 network using successive stretches of history. Eight states take the leading transmitting role. The plotted annotation says the window choice differs from the full-sample choice in 72% of windows. That is not the same statistic as the proportion of adjacent windows in which the identity switches. The adaptive surrogate rule uses past windows and gives about +8% reduction, versus +3% for the fixed full-sample rule and +10% for the oracle-window comparison.

Worked reasoning: Reason through the two kinds of stationarity in the source. Individual incidence series are reported stationary in levels, while the fitted source identity changes over windows. A stable distribution for individual series does not force a time-invariant estimated network role. Conversely, a changing estimated role can reflect both real dynamics and estimation noise. The figure motivates testing adaptation; it does not identify the biological causes of each role change.

Panel b's ordering shows the potential value of updating in this experiment. The oracle's extra advantage is an upper-style reference under privileged window information. It should not be described as a guaranteed realizable policy benefit. The fixed full-sample rule is also distinct from a genuinely prospective rule initialized on an earlier training set.

In ordinary language: A source picked from a long historical average can become outdated. Rechecking past data can improve a decision when the useful source moves. But an update must use information available at the time. A comparison that sees the future is a benchmark, not a deployable forecast rule.

Evidence status: MEASURED_DATED rolling diagnostics plus MODELED_LOCKED control comparisons. The strongest transferable point is the need to test role stability and information timing.

Do not accidentally conclude: The source switches in 72% of adjacent transitions, or an 8% simulated reduction predicts epidemic prevention.

Try it: Why is “differs from the full-sample pick” more precise than “changes 72% of the time”? **Answer:** It states the denominator and reference actually shown. Adjacent-window switching is a different calculation and is not established by that annotation.

D29. Raw-data forecasting is an important but non-significant check

Where we are: Out-of-sample evaluation, page 7; supplementary-reference qualifiers there.

What the passage is doing: The paper separately asks whether an in-sample transmitter helps forecast aggregate downstream stress in held-out observations. This uses real testing data rather than simulated support. Equity shows an 8% forecast-error reduction. Across three stationary-transmitter systems, two have a positive lead and the mean reduction is 3.3%; the three-network sign test gives $p = 0.50$, with a pooled per-unit comparison also non-significant. Migratory-source systems, including epidemic and housing examples, do not retain the same predictive lead.

Worked reasoning: A useful reasoning chain has two separate arrows. Historical data can predict later data. An action can change a system. The first does not establish the second: a predictor might be a marker of a common

driver or something that cannot feasibly be manipulated. Keeping these questions separate protects the empirical forecast check from being oversold as a control trial.

The draft deliberately calls this result suggestive corroboration and places primary weight elsewhere. Do not remove that adjective while retaining the 3.3%. Short macro-series and only three systems limit inference. Pooling many units does not automatically create many independent networks; shared dynamics can couple their errors.

In ordinary language: A model can be tested on observations it did not fit. That is valuable. Here the average gain looks positive, but the evidence is too uncertain to call it a confirmed general improvement. Predicting a future reading is also different from changing that reading through intervention.

Evidence status: MEASURED_DATED held-out predictive evaluation, explicitly non-significant. It is not MODELED_LOCKED intervention evidence and not a field intervention estimate.

Do not accidentally conclude: The paper establishes a 3.3% general forecasting gain, a return on investment or an effect of supporting the transmitter.

Try it: If an earlier variable improves forecasts, what further bridge is needed before targeting it? **Answer:** Evidence that a feasible action on that variable changes relevant downstream outcomes under the intended conditions. Predictive lead alone does not identify that action effect.

D30. Historical-origin recovery and the random-selection null

Where we are: Abstract, page 1; Discussion, page 8; Methods statistics, page 10.

What the passage is doing: Recovery is an exact match between the pre-committed hypothesized historical origin and the diagnosed net transmitter. Five of seven networks match. The source reports a Poisson–binomial probability near 10^{-4} under uniform random node selection, and about 1.5×10^{-3} under a more restrictive comparison involving six networks with transmitter–loudest divergence. Different node counts mean different random-match probabilities, so an ordinary equal-probability binomial calculation would not reproduce the stated null automatically.

Worked reasoning: Reason through the null's dependence on its inputs. It needs a defined candidate set, an origin hypothesis and an exact matching rule for each network. If those are selected after seeing the model, the nominal probability no longer answers the intended pre-commitment question. The draft reports dated written commitments but acknowledges the external timestamp limitation. That limitation remains relevant even when the numerical null probability is small.

The restrictive comparison aims to prevent a trivial explanation in which choosing the loudest node already identifies the source. It addresses a particular alternative, not all plausible non-random predictors. A stronger practical benchmark might use domain-informed alternatives; the source's reported probability should not be relabeled as superiority to every such method.

In ordinary language: Guessing the right node is easier in a small network than a large one. The chance comparison accounts for those different guessing odds. A low chance probability says the recorded matches are unusual under that selection rule. It does not prove every historical account or fitted arrow is correct.

Evidence status: MEASURED_DATED matching outcomes with an ANALYTIC statistical null conditional on recorded hypotheses and candidate sets.

Do not accidentally conclude: A probability near 10^{-4} is the chance that the whole theory is false, or the probability that a real intervention fails.

Try it: What should accompany “five of seven origins recovered”? **Answer:** The exact-match definition, network-specific candidate sets, declared hypotheses, timestamp limitation and null comparator. Without those, a count can sound stronger than the test it actually represents.

D31. Seeds, block bootstrap and small-sample tests answer different questions

Where we are: Discussion, page 8; Methods statistics, page 10; Figure 4 caption, page 15.

What the passage is doing: Main surrogate effect summaries average 16 fixed random seeds shared across controllers. Symmetrization sweeps use 10 seeds. The source reports no data exclusions. Data uncertainty is assessed by a stationary block bootstrap with average block length about √T, using 300 resamples and 95% percentile intervals. Separately, all five prediction-first advantages being positive gives a sign-test $p = 0.031$, while their confirm/refine-versus-falsify partition gives an exact permutation $p = 0.10$. These are different statistics on a small set of domains.

Worked reasoning: Shared seeds make rule comparisons less sensitive to different random simulated conditions. They do not account for all uncertainty in the original observations. A block bootstrap retains stretches of temporal dependence rather than scrambling every row independently. Its output still depends on block length, series length and model assumptions. Wide intervals are therefore part of the result, not a nuisance to hide behind a positive average.

The five positive point estimates can coexist with two failed domain predictions and several individual intervals spanning zero. “Failed” concerns the original structural expectation; “positive” concerns a later numerical advantage. Likewise, $p = 0.031$ and $p = 0.10$ do not contradict one another because they test different summaries.

In ordinary language: Repeating a simulation measures one kind of variation. Resampling the data asks another question about the fitted result. Comparing signs across a handful of systems asks a third. Do not combine those repeats as if they were many independent real-world trials.

Evidence status: MODELED_LOCKED seed summaries; statistical uncertainty conditional on the observed series; SCOPE for inference across five cases.

Do not accidentally conclude: Sixteen seeds are sixteen independent datasets, or every case is individually significant because an aggregate sign test is small.

Try it: Which seed count belongs to the symmetrization sweeps? **Answer:** Ten, not the sixteen used for main effect summaries. Keeping those budgets distinct is necessary for reproducing the reported procedures.

Interpretation, limits and the research record

D32. The geometric motivation is not the empirical result

Where we are: Introduction closing, pages 2–3; geometric interpretation, page 8.

What the passage is doing: The draft describes Last Straw Analytics as a conceptual motivation: interference resembles leakage through a unit's Markov blanket, and a cascade resembles loss of a separating boundary. It refers to non-Hermitian spectra, optimal transport on a learned latent manifold and

discrete-Morse descriptions in supplementary work, within a free-energy/active-inference idiom. It explicitly says this apparatus is demonstrated chiefly on synthetic constructions and a limited empirical signature; the main empirical pipeline does not depend on it.

Worked reasoning: A Markov blanket is a specified set of variables that screens a system from other variables in a probabilistic model. It is not automatically a literal wall, a cell membrane or a verified municipal boundary. A learned manifold is a representation of data, not proof that the real system occupies a unique physical surface. The source does not supply enough detail in this PDF for this companion to derive all of the supplementary geometric machinery.

The correct teaching task is therefore to explain the role of these ideas and point to their evidence limit. Introducing additional equations, fitted free energies or claimed phase boundaries would create a different document and risk promoting interpretation into analysis. The paper's own distinction is a strength worth retaining.

In ordinary language: A picture can help organize a research question without being a tested law. Here the picture is a protective boundary that stops separating a unit from its surroundings. The controller results stand on the fitted time-series model and its tests. They do not validate every part of that broader picture.

Evidence status: INTERPRETIVE motivation; SCOPE for unprinted and chiefly synthetic developments. No new geometric derivation or validation is claimed here.

Do not accidentally conclude: Success of a VAR-based controller validates a universal geometry of collapse, or that the term Markov blanket identifies a measured protective boundary in each dataset.

Try it: Can you explain the main findings without mentioning learned manifolds? **Answer:** Yes. Calibrate the surrogate, diagnose net predictive roles, compare controllers and perturb asymmetry. That independence is explicitly part of the source's argument.

D33. Short series, volatility and uncertainty limit the diagnosis

Where we are: Limitations, pages 8–9; Methods horizon and stationarity notes, page 9.

What the passage is doing: The draft acknowledges short time series, fragile rankings, wide controller-advantage intervals and possible distortions from variance spillovers. It reports horizon stability in the studied networks but calls for formal causal-invariance checks, especially in equity volatility settings. Nonlinear short-series directed-dependence estimators are proposed as future alternatives, not analyses already carried out in this paper.

Worked reasoning: Reason through two uncertainty sources. Simulation variation comes from the random cascade experiment. Estimation variation comes from fitting a network to a finite, dependent record. A controller can be consistently good on one fitted matrix yet have an uncertain advantage across plausible fitted matrices. That is why shared seeds and many simulated steps cannot replace resampling or other checks of the underlying data model.

The source states that only equity and held-out COVID-19 individually exclude zero in block-bootstrap advantage intervals. Teaching the other point estimates as established standalone effects would remove the principal limitation. It is possible to discuss their structural pattern while saying their individual magnitudes are imprecise.

In ordinary language: A stable-looking answer from one fit may still be uncertain. Limited data, common shocks and changing variability can affect which node leads the ranking. Check how much the answer moves when data or sensible choices change. A suggested future method is not evidence that the current result has already passed that test.

Evidence status: SCOPE and reported robustness analysis; OPEN future work on alternative estimators and causal invariance. The guide does not claim those future checks are completed.

Do not accidentally conclude: Ridge and non-negativity eliminate model uncertainty, or that a positive point estimate settles the sign of a real intervention effect.

Try it: A fitted network gives the same leader for several horizons but changes leader under bootstrap resampling. Are the checks inconsistent? **Answer:** No. One varies a forecast horizon on the fitted model; the other varies the sampled evidence used to fit it. They probe different sensitivities.

D34. Persistence, scale and the outlook must stay bounded

Where we are: Limitations and Outlook, page 9.

What the passage is doing: The currency network's non-stationary levels are disclosed as a specific limitation. Near-unit-root spectral radii are interpreted primarily as persistence, not certified supercritical-to-subcritical transitions. Computations are CPU-only at moderate sizes, $N \leq 400$. GPU expansion is a future design target. The outlook proposes transfer across directed diffusion systems with quiet origins distinct from loud casualties, while the evidence remains conditional on the tested diagnostic and surrogate framework.

Worked reasoning: A worked transfer plan starts with the new domain's observations and decision target. Check whether units, time bins and a reinforcing linear model make sense. Examine stationarity and role uncertainty. Compare source-aware and reactive rules under a declared, feasible model intervention. Only then consider whether scaling the computation is the real bottleneck. More hardware cannot repair an inappropriate estimand or a missing causal bridge.

The phrase “lower-tier diffusion systems” in the outlook is a qualitative scope description, not a new numerical classifier supplied by the draft. A reader should not invent a cutoff or certify a new infrastructure system as eligible from that phrase alone. The operational validity map is an empirical pattern that requires testing in each new application.

In ordinary language: A large persistence score does not prove a system is about to cross a physical tipping point. Nor does a fast experiment on a moderate network show that the code has been tested at massive scale. Separate what ran from what the author hopes to build next.

Evidence status: SCOPE for demonstrated scale and statistical interpretation; OPEN for GPU and broader deployment.

Do not accidentally conclude: All analyses involved 400 nodes, that GPU results exist, or that a spectral radius near one forecasts a real collapse.

Try it: What is the first reason a very large sensor network might fail to support this workflow? **Answer:** It may lack suitable, aligned, sufficiently informative observations or a defensible model/intervention definition. Computational size is only one issue; the source does not claim a universal scalable solution.

D35. Availability statements are a reproducibility map, not a rerun certificate

Where we are: Availability of data and materials, page 11; Methods reproducibility, page 10.

What the passage is doing: The paper lists public aggregate sources: Johns Hopkins COVID-19, CDC ILINet via Delphi Epidata, FRED exchange rates, EPA AirData PM2.5, BTS flight records, UCDP GED v24.1 and eight national equity indices. It points to the GitHub project and frozen software/data DOI 10.5281/zenodo.21211653, with retrieval scripts, processed series, pinned environment, seeds and per-figure regeneration. These are the paper's availability claims, not a fresh full reproduction performed for this companion.

Worked reasoning: Reason through source versus derivative. Daily measurements and original index records are upstream data. Weekly closing levels and aligned matrices are processed inputs. Fitted networks, controller results and figures are downstream outputs. Reproducing a figure needs the correct chain, not merely a file with a plausible name. The paper says its archive contains this chain; users should consult the current root README for maintained commands and explicit figure inputs.

The archive DOI identifies software/data, not an assigned manuscript preprint DOI. The selected author draft is linked separately. The later archive timestamp also cannot prove the initial predictions were written before analysis. Availability and chronology answer different questions.

In ordinary language: A good research record tells you which inputs and code produced each output. A public source name alone is not enough: you also need dates, versions, transforms and the exact analysis path. The archive helps preserve that path. The maintained repository can change after the paper's snapshot.

Evidence status: SCOPE and reported reproducibility provisions. This teaching preparation inspected the source, not every remote dataset or every full scientific computation.

Do not accidentally conclude: Reading the code archive certifies all reported outcomes, or that a mutable latest checkout is byte-identical to the frozen analysis snapshot.

Try it: What should a reproducibility note distinguish? **Answer:** Exact manuscript, software snapshot, input vintages, derived data, parameters and output. If any are missing, name the gap rather than calling the whole chain reproduced.

D36. Declarations and references tell you who is responsible and what supports what

Where we are: Declarations, page 11; References 1–29, pages 12–13; corresponding-author note, page 1.

What the passage is doing: The author reports independent research, no external funding or competing interests, sole authorship and use of public aggregate data. Ethics and publication-consent statements are marked not applicable for this study. AI tools assisted literature search, coding, drafting and checks; responsibility for scientific decisions and verification remains with the author. The source says AI neither generated nor altered the real datasets, that the human author wrote the prediction records, and that no AI system is an author. It thanks the public-data maintainers. These declarations are research-record statements, not statistical evidence for the regularity.

Worked reasoning: The bibliography has several jobs. Rubin and interference papers supply causal background; Diebold–Yılmaz and Granger supply predictive-direction methods; control and source-identification works supply alternatives; UCDP supplies a dataset; geometric references motivate the interpretation; short-series and volatility papers motivate limitations. Later tail-risk work supplies convergent

descriptive context, not an independent replication of this control experiment. All 29 entries remain in the source bibliography; this companion explains their roles rather than reproducing a second literature review.

No new claim about ethics approval, authorship or completed verification is made by paraphrasing the declarations. In particular, the source's AI-assisted verification statements are attributed to the source. They are not evidence that this teaching document has passed a full manuscript harness or a human learning test.

In ordinary language: Knowing who did the work and how tools were used helps you judge accountability. It does not make an effect true or false. Likewise, a reference can justify a method or provide context without directly confirming this paper's result. Read the role of the citation, not just its presence.

Evidence status: SCOPE for administrative declarations; INTERPRETIVE organization of reference roles. Neither is an added empirical finding.

Do not accidentally conclude: An archive, an AI disclosure or an author's approval substitutes for independent reproduction or human educational validation.

Try it: Does a citation to physical power-grid influence establish physical links in the equity model? **Answer:** No. It supplies a different-domain analogy about influence versus connectivity. Each dataset still needs its own measurement and model justification.

Glossary and notation

Term	Meaning in this paper
Node/unit	One included state, airport, currency, market or other aggregate series.
Stress series	The declared numerical outcome modeled over time; not a universal measure of harm.
Interference	One unit's outcome depends on other units or their treatment.
SUTVA	A causal-inference assumption whose no-interference part excludes such cross-unit treatment dependence.
VAR(1)	Linear multivariate prediction using the previous time step.
Φ	Fitted coefficient matrix; entries are constrained non-negative here.
Residual/innovation	Variation left unexplained by the fitted prediction.
Ridge	Penalty that shrinks coefficients to stabilize estimation.
Stationarity	Stability of relevant distributional behavior over time; not a constant observed value.
GFEVD	Model-based allocation of forecast-error variance across units' innovations.
Connectedness	Summary of that variance sharing, not a physical edge count.
Net transmitter	Unit maximizing outward minus inward connectedness.

Term	Meaning in this paper
Loudest unit	Unit with greatest realized stress under the declared mean/cumulative summary.
Surrogate	Simplified fitted stand-in for the original system.
Spectral radius ρ	Largest absolute eigenvalue; helps describe growth in the linear dynamics.
Supercritical regime	Deliberately amplifying model setting used for cascade contests here.
Interdiction	Simulated allocation of support to attenuate the modeled cascade.
Cascade reduction	Improvement relative to the no-action surrogate; negative means worse under that objective.
Symmetrization	Interpolation toward the symmetric matrix projection, followed by radius adjustment.
α	Interpolation parameter; one is original and zero is symmetric before rescaling.
Static targeting	Choose a source once for the specified contest.
MPC	Model predictive control; repeatedly plan support using model forecasts.
Oracle	Idealized comparison with privileged information, not necessarily deployable.
Block bootstrap	Resampling stretches of time series to retain some dependence.
Prediction-first	Written prior expectation without verified external pre-analysis timestamp for the original five tests.
Poisson–binomial null	Sum of independent Bernoulli indicators with potentially different success probabilities, used for the stated random-match comparison.
Markov blanket	Variables that screen a system from others in a specified probabilistic model; an interpretive analogy here.

Twelve-item transfer assessment with answers

Write answers before reading the key. This is an unadministered self-study exercise, not measured educational effectiveness.

Core questions

1. Trace the three stages in Figure 1 and identify where the intervention happens.
2. Define net transmitter and loudest unit. Why can they differ?
3. Use S and A to explain equation 3, including the subsequent radius adjustment.
4. Why compare MPC with a fixed source-aware rule as well as reactive triage?

Boundary questions

5 Does a fitted directed arrow identify an intervention effect?

6 Why are the five original tests prediction-first rather than externally preregistered?

7 Can the smoke reduction or the 3.3% raw-data forecast gain be called a confirmed field benefit?

8 Does equation 2 establish horizon-invariant ranks for every possible network?

Transfer questions

9 A new network has high connectedness but near symmetry. What should be tested before promising a source-targeting benefit?

10 A transmitter is also the loudest node. What does a tie between the two rules mean?

11 A rolling-window leader differs from the full-sample leader in many windows. What information-timing comparison is needed?

12 A sensor system has strong trends, little history and unverified links. What should transfer first from this paper?

Answer key

1 Calibrate from observed series, diagnose predictive roles, then simulate interdiction on the rescaled surrogate. The last stage is not a real intervention.

2 The first maximizes outward-minus-inward forecast-variance sharing; the second maximizes the declared realized-stress summary. Role and local magnitude are different quantities.

3 $\Phi = S+A$, so the blend is $S+\alpha A$ before rescaling. Each blend is then multiplied to reach the common radius. Do not claim the final symmetric entries remain numerically identical.

4 It tests whether repeated planning adds value beyond a good diagnosis. Beating only a weak reactive rule would not establish that extra machinery is needed.

5 No. Predictive dependence can reflect omitted drivers and model choices; feasible action effects need another evidential bridge.

6 The later archive verifies existence by deposit, not an independently verified timestamp before each analysis. Later OSF registrations concern other, unanalyzed systems.

7 No. Smoke is an uncertain surrogate effect; the raw-data mean is non-significant and concerns forecasting, not intervention.

8 No. The draft reports agreement in its evaluated networks. A new system still needs a specified horizon target and sensitivity checks.

9 Diagnose role asymmetry and gap, run a matched-budget contest and relevant perturbation, and quantify uncertainty. High total sharing alone is insufficient.

10 The targets may coincide, leaving little comparison. This is distinct from weak coupling or near symmetry and does not deny that direction exists.

11 Compare realizable past-only updating with a stated fixed baseline. Keep any oracle using privileged information separate; do not confuse disagreement with adjacent-window switching.

12 Transfer the questions and diagnostic workflow, not a certified ranking or expected benefit. Check transformations, sampling, model suitability, uncertainty and intervention meaning first.

For a one-sentence teach-back, state the allocation contrast, its surrogate evidence, its principal causal limitation and one test needed in a new domain. A human reader's clarity and transfer scores are unrecorded.

Preparation and source record

The source PDF above remains unchanged. All paper-result numbers are attributed to that exact draft. Explicit hypothetical arithmetic in D04 illustrates definitions only; symbolic reasoning does not add empirical observations. The selected PDF's supplementary references are explained at the level printed, not reconstructed as nonexistent analyses in this document.

The concise guide was preserved separately. The older directedness learning bundle informed its initial teaching sequence, but this complete companion was developed against the exact July draft and its six figures. Source comparisons, arithmetic checks and automated structural checks are AI-assisted. No full scientific rerun, human audience review, accessibility signoff, administered transfer assessment or full G14 completion is claimed. PDF rendering and derivative checks are recorded separately during release preparation.